group head, Muthunayagam's brief was to develop propulsion systems with the propellant developed and delivered by the propellant group headed by Gowarikar.

Muthunayagam exceeded his brief, and started developing propellants on his own. His reason, that Gowarikar's group was not delivering good propellants, and that he was forced to experiment with his own, though with ample reason, infuriated Gowarikar. All these, for the record, were solid propellants for the minion Rohini-75, which did virtually nothing but rise to the sky for a few kilometers and plunged into the Arabian Sea. Satellite launches were still a distant dream, and my occasional prodding for liquid propulsion was seen as an ambition bordering on absurdity.

Gowrikar's group was not sitting idle; they were trying to develop several high-energy propellants, but just could not get them right. Taking this as a reason, and evidently to belittle Gowarikar, Muthunayagam told me that our group has to acquire cordite, a low-energy propellant made by the Cordite Factory at Aravankadu in the Nilgiris, to test-fire Rohini.

Amidst these efforts, one man felt like a king—M R Kurup, head of the Rocket Propellant Plant. Kurup's team was making PVC-based propellant for Centaur rockets supplied by the French as part of a deal Sarabhai struck in the late 1960s. The French were supplying the rockets and the technology to make PVC-based propellant, till India could make its own. So, Kurup was delivering something which was being handed out by the French, while the groups headed by Gowarikar and Muthunayagam were struggling to develop an indigenous technology to make propellants.

As part of the Centaur deal, Eswar Das was in charge of fabrication, and Abdul Kalam's job was to integrate and marry the payloads made by Physical Research Laboratory, Ahmedabad with the rocket. With nothing much to do on this front, Kalam, who always enjoyed a free hand, started working on a rocket, which he called D1. When he told me about the experiment, which was nothing more than the existing Rohini rocket, with the same 75 mm diameter, Kalam said he needed to do something to keep him occupied.

'Why the name D1?' I asked. 'Have you made A1, B1 and C1?'